PHY-112 | Principles of Physics-2

Akiful Islam (AZW)
Spring 2025 | Class #9

Department of Mathematics & Natural Sciences

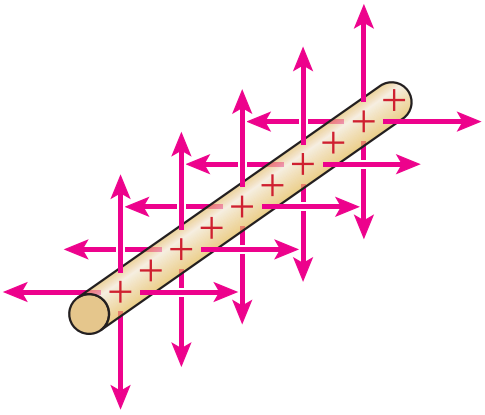
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Potentials of Continuous Sources

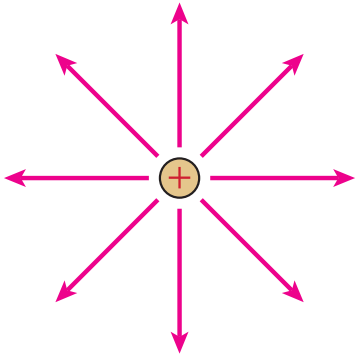
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Recall the Line Charge

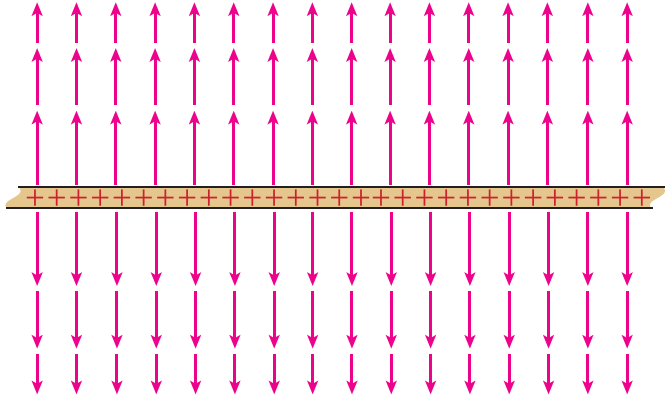
Infinitely long line charge



3D view



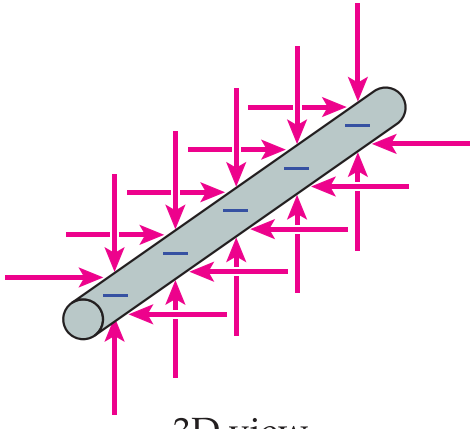
End view



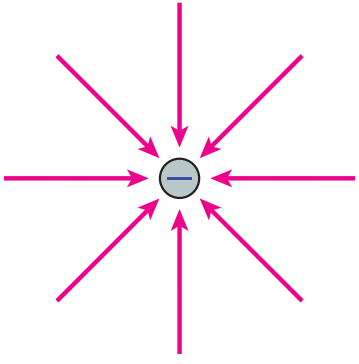
Side view

$$\vec{E}_{\text{Line}} = \left(\frac{\lambda}{2\pi\epsilon_0 r}\right)\hat{r}$$

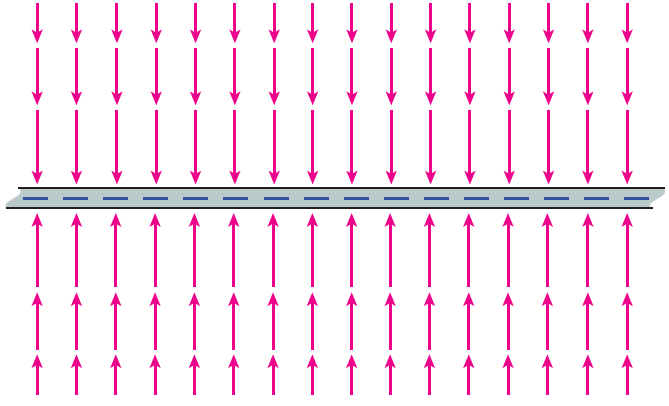
$$\lambda = \frac{Q}{L}$$



3D view



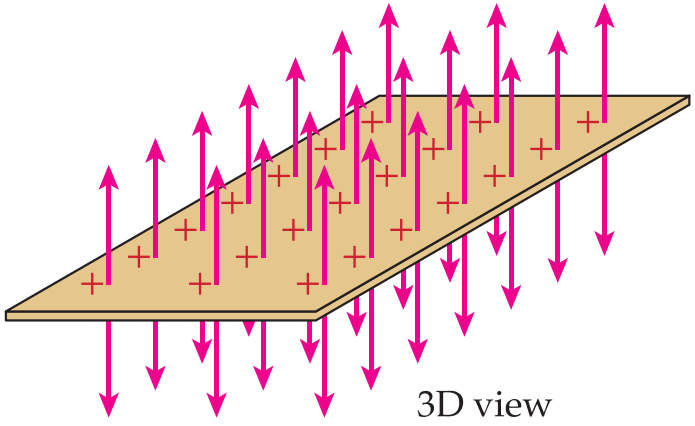
End view



Side view

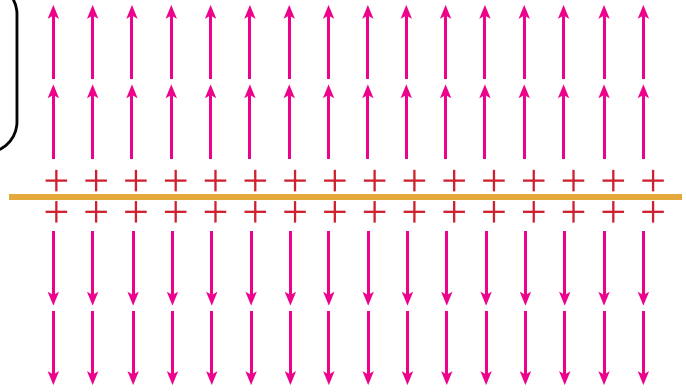
Recall the Plane/Surface Charge

Infinitely extended surface charge



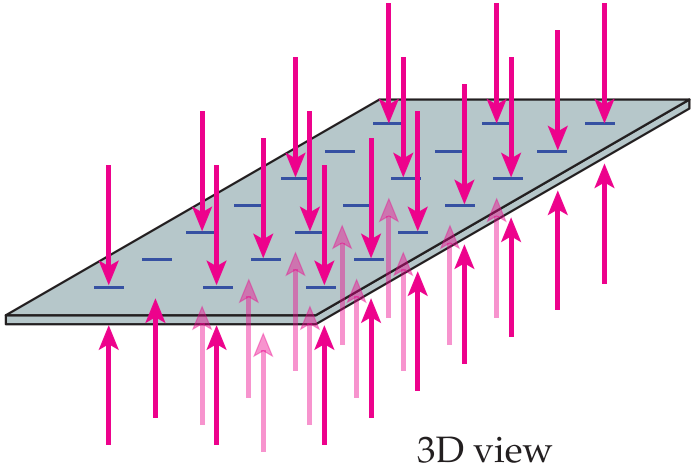
3D view

$$\sigma = \frac{Q}{A}$$

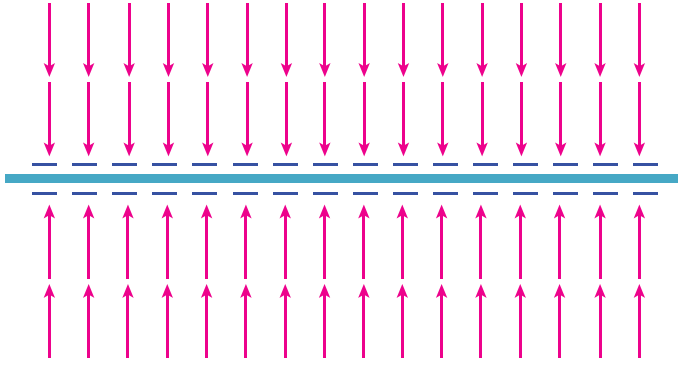


End view = Side view

$$\vec{E}_{\text{Sheet}} = \left(\frac{\sigma}{2\epsilon_0}\right)\hat{n}$$



3D view



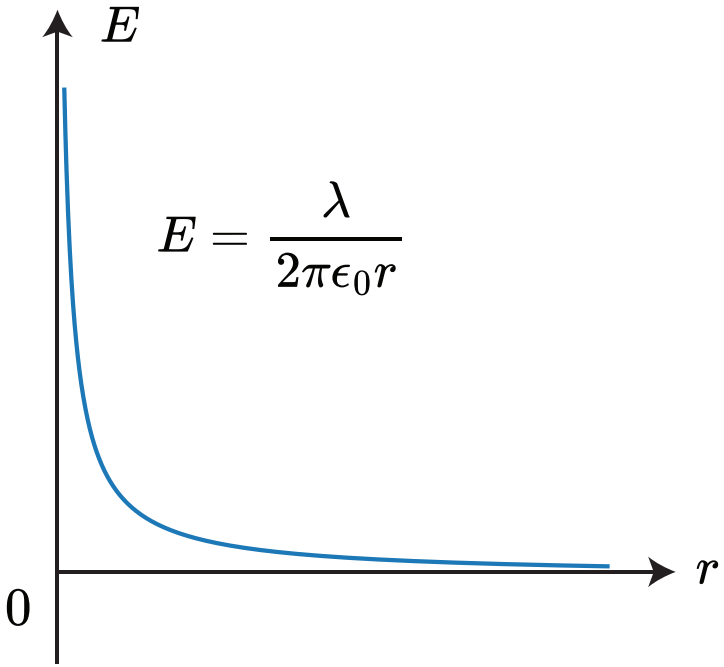
End view = Side view

Potentials of 3 Key $\vec{E}$ Field Sources: Continuous
Remember Line, Plane and Volume Charges

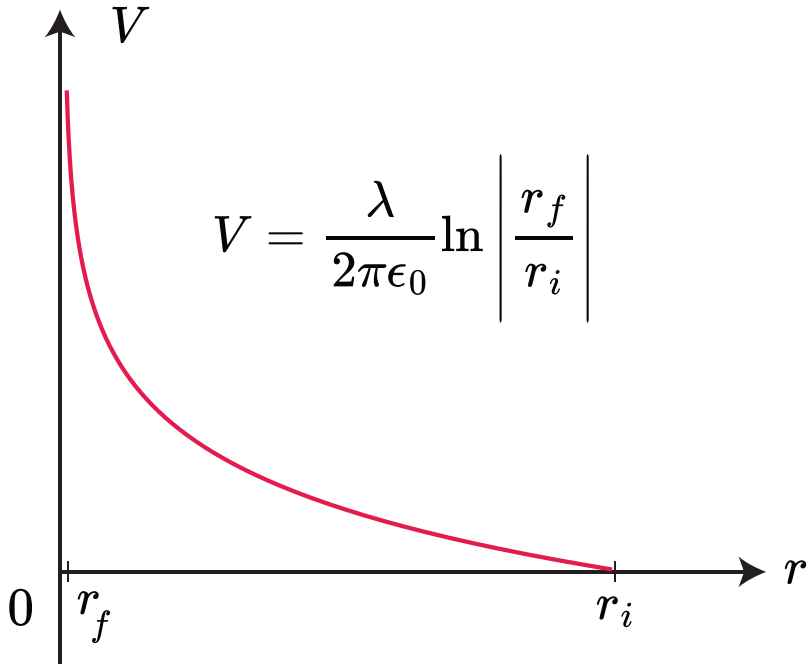
Line	Surface
$\vec{E} = \left(\frac{\lambda}{2\pi\epsilon_0 r}\right)\hat{r}$	$\vec{E} = \left(\frac{\sigma}{2\epsilon_0}\right)\hat{n}$
$V = -\int_{+}^{-} \vec{E} \cdot d\vec{l}$	$V = -\int_{+}^{-} \vec{E} \cdot d\vec{l}$
$V = -\left(\frac{\lambda}{2\pi\epsilon_0}\right)\int_{+}^{-} \frac{1}{r} dr$	$V = -\left(\frac{\sigma}{2\epsilon_0}\right)\int_{+}^{-} dr$
$V = -\left(\frac{\lambda}{2\pi\epsilon_0}\right)[\ln r]_i^f$	$V = -\left(\frac{\sigma}{2\epsilon_0}\right)[r]_i^f$
$V = \frac{\lambda}{2\pi\epsilon_0}\ln\left\lvert\frac{r_i}{r_f}\right\rvert$	$V = -\left(\frac{\sigma}{2\epsilon_0}\right)\Delta r$

Plot the behavior of E and V

Infinitely long line charge



$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$



$$V = \frac{\lambda}{2\pi\epsilon_0} \ln \left| \frac{r_f}{r_i} \right|$$

Incepting Ideas (1)

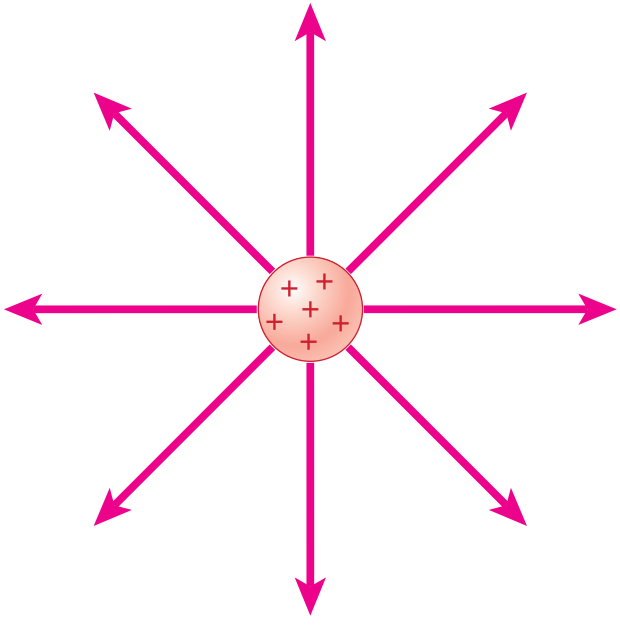
Hint: The *infinity* is actually at a finite distance

Q1: A very long insulating cylindrical shell with a radius of 6.70 cm carries a charge of linear density of 9.00 nC m^{-1} spread uniformly over its outer surface. What would a voltmeter read if it were connected between (a) the surface of the cylinder and a point 4.70 cm above the surface and (b) the surface and a point 1.00 cm from the central axis of the cylinder?

Q2: A very long insulating cylinder of charge of radius 2.50 cm carries a uniform linear density of 18.0 nC m^{-1}. If you put one probe of a voltmeter at the surface, how far from the surface must the other probe be placed so that the voltmeter reads 190 V?

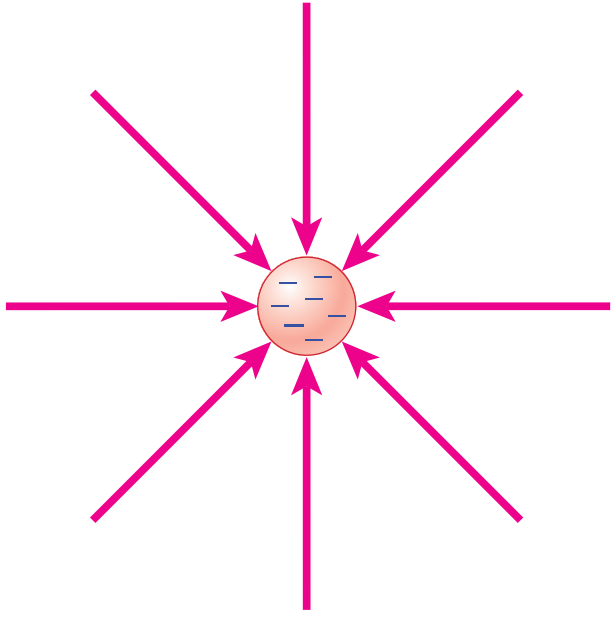
Recall the Volume Charge

Spherically symmetric charge distribution



2D view

$$\rho = \frac{Q}{V}$$

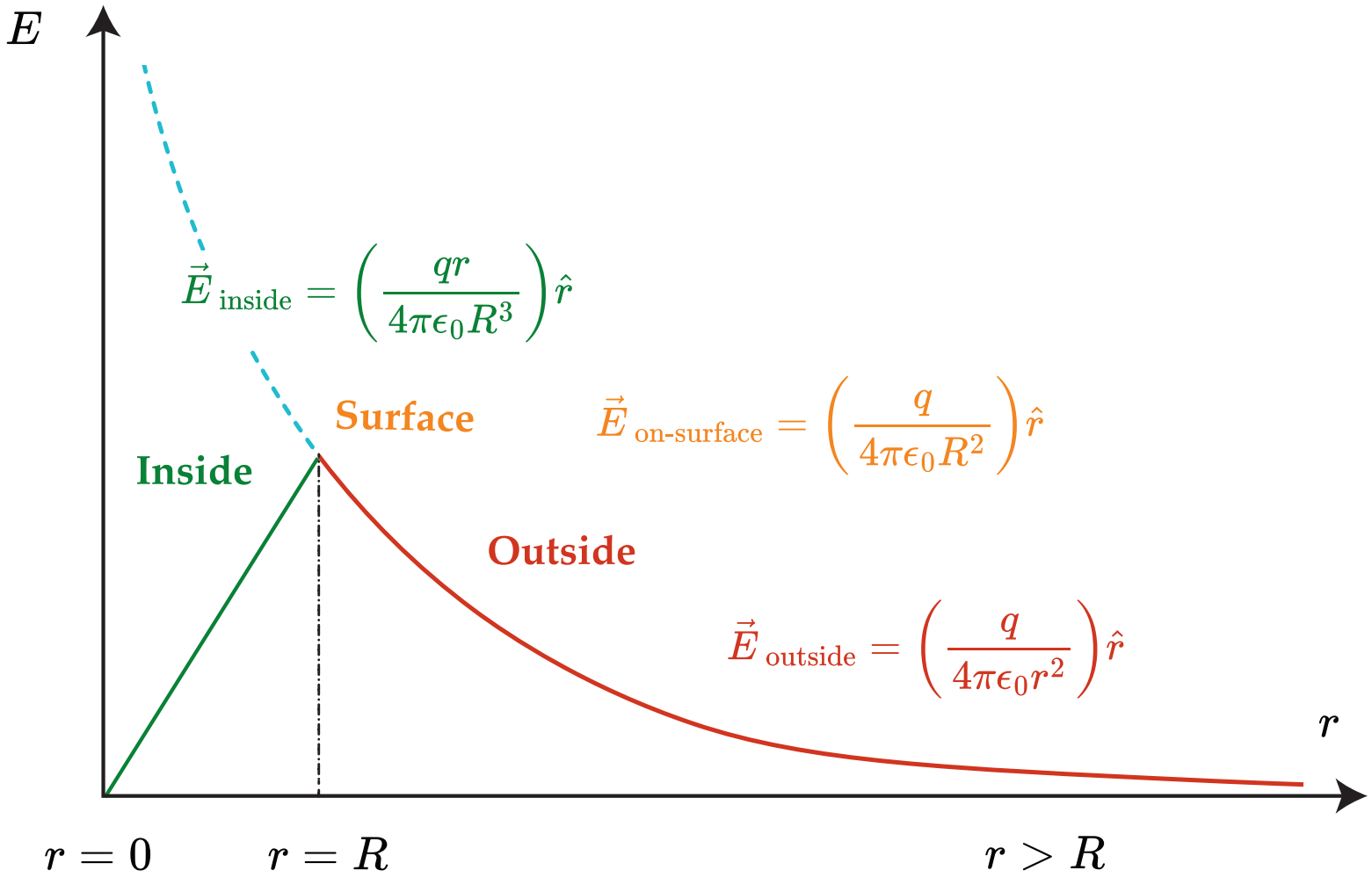


2D view

Field acts radially outward (positive), or inward (negative)

Electric Field of a 3D Volume Charge

Spherically symmetric charge distribution



Potential of a 3D spherical Charge

We approach the 3D charge from infinity

$$\Delta V = V_f - \underbrace{V_\infty}_{0} = -\int_\infty^{r<R} E dr'$$

$$= -\int_\infty^{R} E_{\text{out}} dr' - \int_R^{r<R} E_{\text{in}} dr'$$

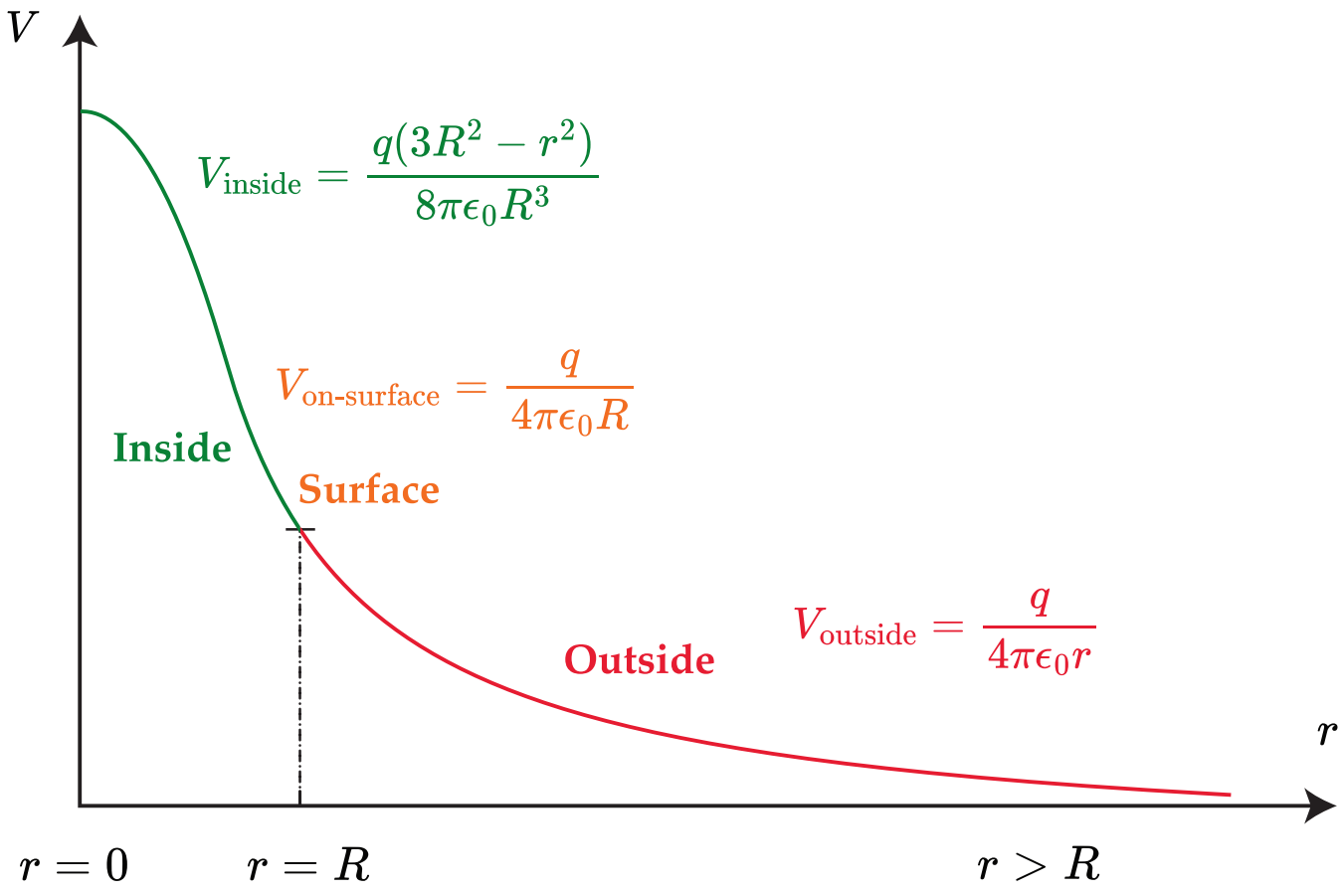
$$= -\frac{q}{4\pi\epsilon_0}\int_\infty^{R} \frac{1}{r'^2} dr' - \frac{q}{4\pi\epsilon_0 R^3}\int_R^{r<R} r' dr'$$

$$= \frac{q}{4\pi\epsilon_0 R} + \frac{q}{8\pi\epsilon_0 R^3}(R^2 - r^2)$$

$$= \frac{q(3R^2 - r^2)}{8\pi\epsilon_0 R^3}$$

Potential of a 3D spherical Charge

We approach the 3D charge from infinity



Incepting Ideas (2)

Hint: Conducting sphere holds no charge inside it

Q: What are (i) the charge and (ii) the charge density on the surface of a conducting sphere of radius 0.15 m whose Potential is 200 V (with $V = 0$ at infinity)?

Incepting Ideas (3)

Hint: Field is zero inside, but not potential

Q: An empty hollow metal sphere has a potential of $+400$ V with respect to ground (defined to be at $V = 0$) and has a charge of 5.0×10^{-9} C. Find the electric Potential at the center of the sphere.

Incepting Ideas (4)

Hint: Do a run from inside out with V

Q: A Gaussian sphere of radius 5.0 cm is centered on a ball that has a radius of 10.0 cm and a uniform charge distribution. The total (net) electric flux through the surface of the Gaussian sphere is $+5.60 \times 10^4$ N m^2 C^{-1}. What is the electric potential at (i) $r = 5.0$ cm, (ii) $r = 10.0$ cm, (iii) $r = 15.0$ cm from the center of the ball?

Potential for 3 Continuous Sources

Glance over it at once

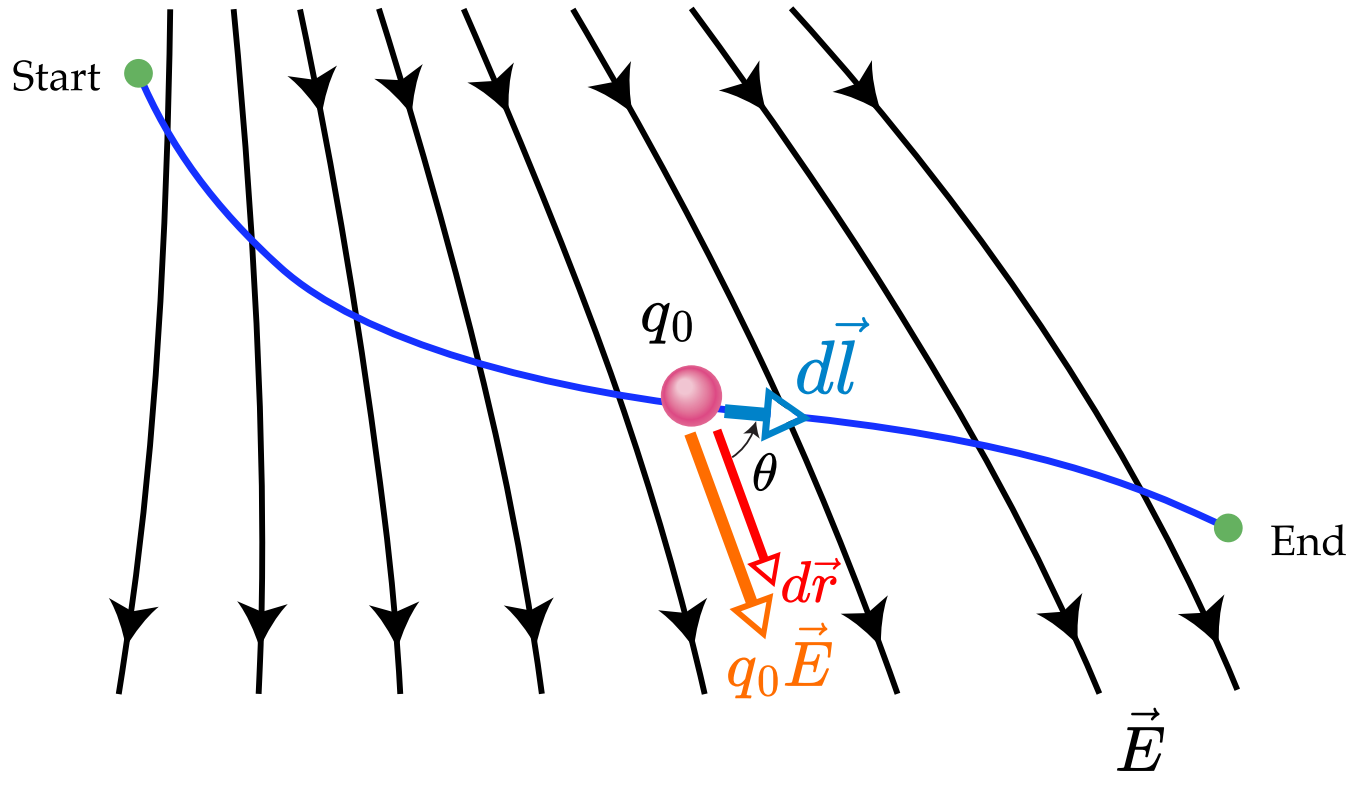
- Line Charge $\longrightarrow V = -\left(\frac{\lambda}{2\pi\epsilon_0}\right) \ln\left|\frac{r_f}{r_i}\right|.$
- Surface Charge $\longrightarrow V = -\left(\frac{\sigma}{2\epsilon_0}\right)(r_f - r_i)$
- Volume Charge $\longrightarrow V = \begin{cases} \frac{q(3R^2 - r^2)}{8\pi\epsilon_0 R^3} & \text{(Inside)} \\ \frac{q}{4\pi\epsilon_0 R} & \text{(On Surface)} \\ \frac{q}{4\pi\epsilon_0 r} & \text{(Outside)} \end{cases}$

Potentials of Uniform sources

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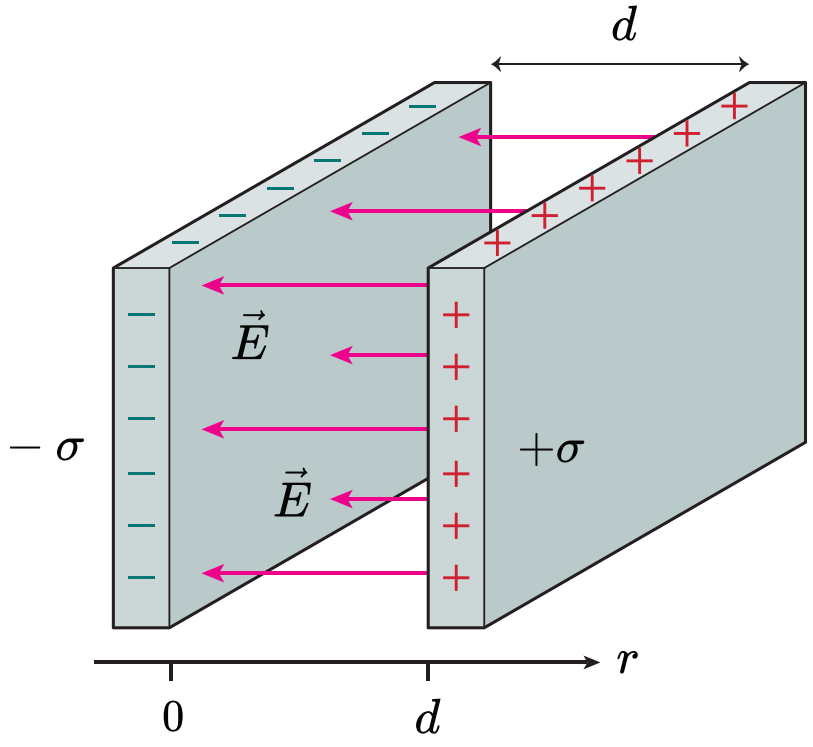
Potential Difference in Electric Fields

Between any two points



Potential Change in a uniform Electric Field

Example: A parallel-plate capacitor

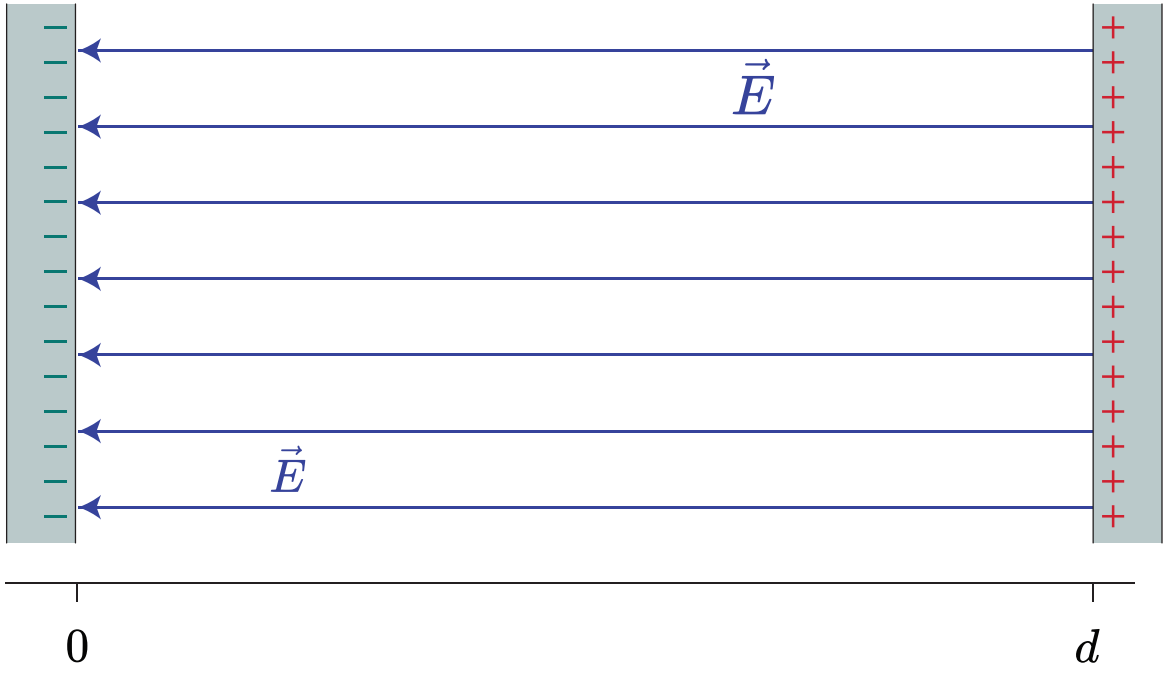


$$\Delta V = -\int_{+}^{-} \vec{E} \cdot d\vec{l} = -\int_{+}^{-} E dr = -E\Delta r.$$

Incepting Ideas (5)

Hint: Potential does not care about the observers

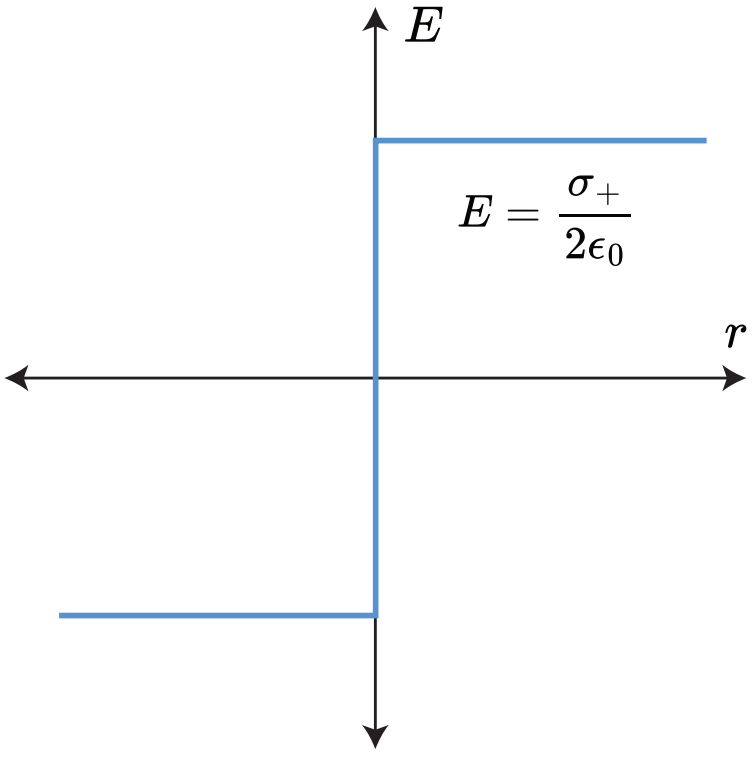
Q: A parallel-plate capacitor with a 2.0 mm spacing produces a uniform $\vec{E}$-field of intensity $2.85 \times 10^5\ \mathrm{N\,C^{-1}}$.



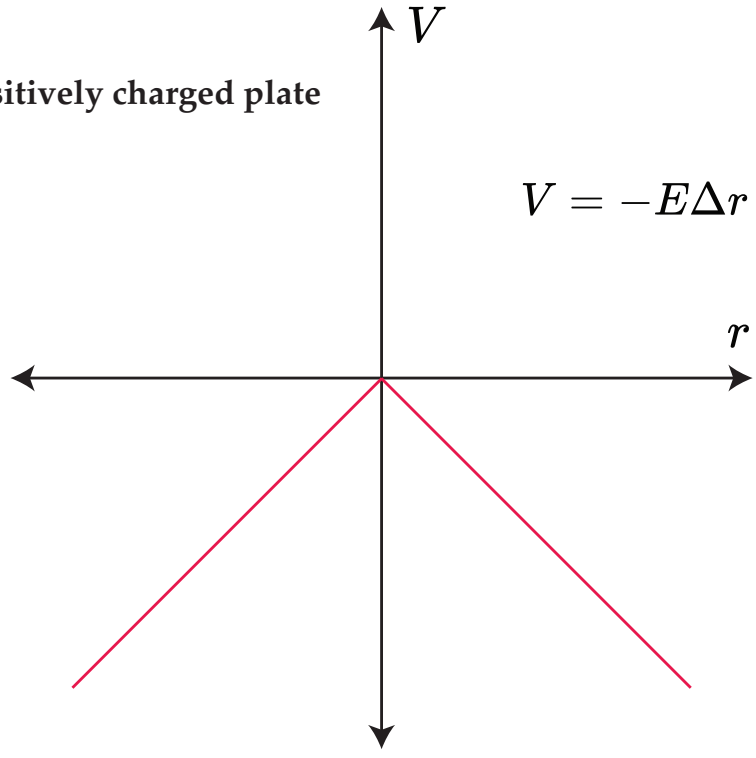
- What is the net potential difference between the plates?

Plot the behavior of E and V

Single Positive Plate: Watch the signs

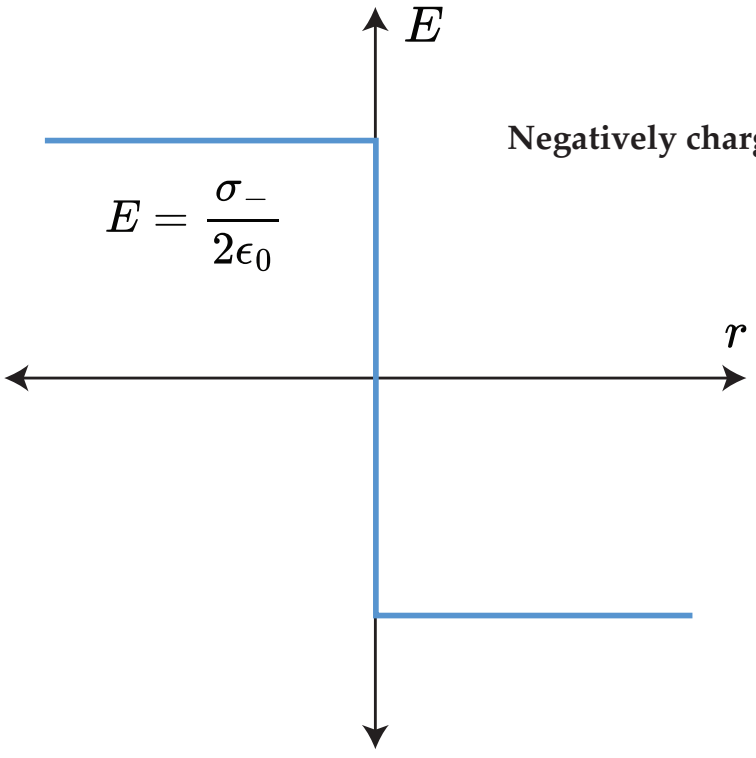


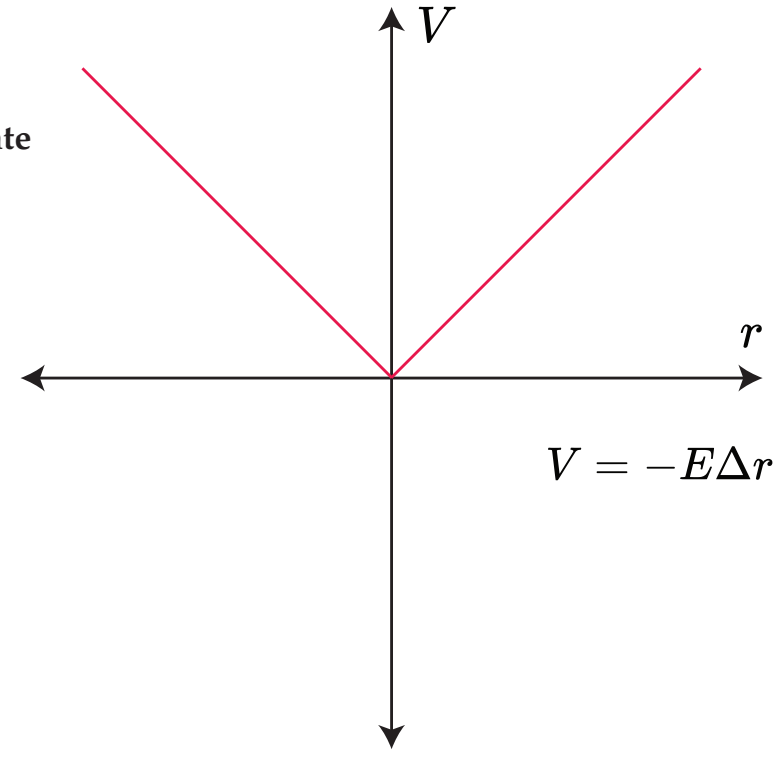
Positively charged plate



Plot the behavior of E and V

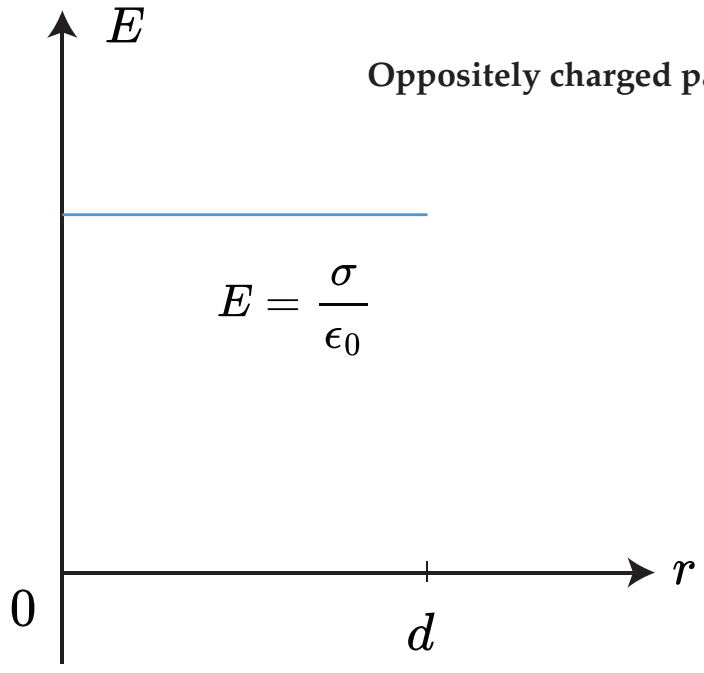
Single Negative Plate: Watch the signs

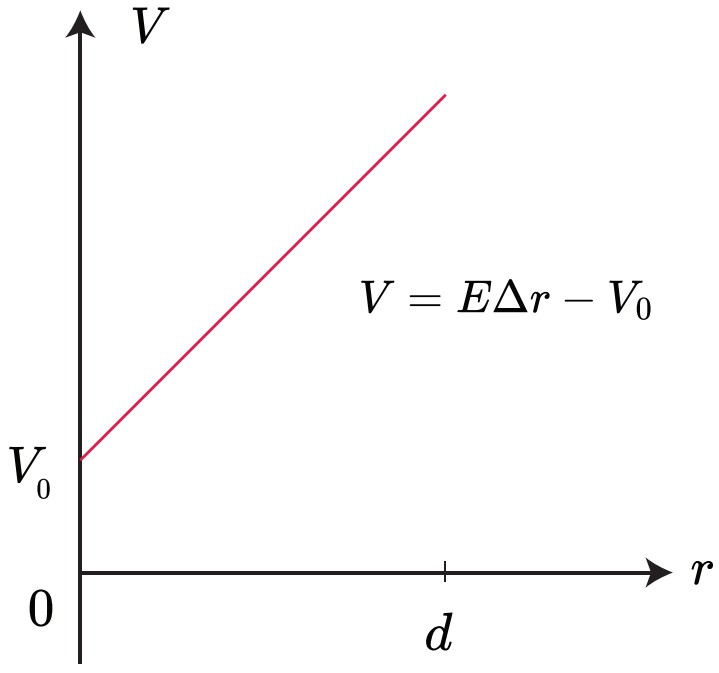




Plot the behavior of E and V

Parallel Plates: Watch the signs





Incepting Ideas (6)

Hint: Constant field, but variable potential

Q: The electric potential V in the space between two flat parallel plates 1 and 2 is given (in volts) by $V = 1500x^2$, where x (in m) is the perpendicular distance from plate 1. At $x = 1.3$ cm, (a) what is the magnitude of the electric field, and (b) is the field directed toward or away from plate 1?

That is it for today!

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